Statistical Inference and Multivariate Analysis (MA324)

Lecture Slides

Topic 01

Delta Method and Bootstrap

Jan - May, 2026

Section 1

Delta Method

Introduction & Intuition

- **The Problem:** We know the asymptotic distribution of an estimator $\hat{\theta}$, but we want to know the asymptotic distribution of a continuous function $g(\hat{\theta})$.
- **The Solution:** The Delta Method uses a first-order Taylor series expansion to approximate the variance of the transformed estimator.
- **Taylor Expansion:** Around the true parameter θ, the function $g(\hat{\theta})$ can be approximated as:

$$g(\hat{\theta}) \approx g(\theta) + g'(\theta)(\hat{\theta} - \theta)$$

- Rearranging terms, we get:

$$g(\hat{\theta}) - g(\theta) \approx g'(\theta)(\hat{\theta} - \theta)$$

The Univariate Delta Method

Let X_n be a sequence of random variables such that:

$$\sqrt{n}(X_n - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$$

Suppose g is a function such that its derivative $g'(\theta)$ exists and is non-zero. Then:

$$\sqrt{n}(g(X_n) - g(\theta)) \xrightarrow{d} \mathcal{N}\left(0, \sigma^2 \left[g'(\theta)\right]^2\right)$$

Implication: The variance of the transformed estimator is approximately the original variance multiplied by the square of the derivative of the transformation evaluated at θ.

Proof Sketch: Univariate Delta Method

- By Taylor's Theorem and the Mean Value Theorem, we can write:

$$g(X_n) = g(\theta) + g'(\tilde{\theta})(X_n - \theta)$$

  where $\tilde{\theta}$ lies strictly between X_n and θ.
- Rearranging and multiplying by $\sqrt{n}$:

$$\sqrt{n}(g(X_n) - g(\theta)) = g'(\tilde{\theta})\sqrt{n}(X_n - \theta)$$

- Since $\sqrt{n}(X_n - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$, we know $X_n \xrightarrow{p} \theta$. Thus, $\tilde{\theta} \xrightarrow{p} \theta$.
- By the Continuous Mapping Theorem, $g'(\tilde{\theta}) \xrightarrow{p} g'(\theta)$.
- Finally, applying **Slutsky's Theorem**:

$$g'(\tilde{\theta})\sqrt{n}(X_n - \theta) \xrightarrow{d} g'(\theta) \cdot \mathcal{N}(0, \sigma^2) \equiv \mathcal{N}\left(0, \sigma^2[g'(\theta)]^2\right)$$

Example: Estimating Survival Probability

- Let lifetimes $T_1, \ldots, T_n \sim \text{Exp}(\lambda)$. The MLE for the rate is $\hat{\lambda} = 1/\bar{T}$.
- From asymptotic likelihood theory, we know:

$$\sqrt{n}(\hat{\lambda} - \lambda) \xrightarrow{d} \mathcal{N}(0, \lambda^2)$$

- We want the asymptotic distribution of the survival function at time t_0: $S(t_0) = \exp(-\lambda t_0)$.
- Let $g(\lambda) = \exp(-\lambda t_0)$. The derivative is $g'(\lambda) = -t_0 \exp(-\lambda t_0)$.
- Applying the Delta Method:

$$\sqrt{n}\left(\exp(-\hat{\lambda} t_0) - \exp(-\lambda t_0)\right) \xrightarrow{d} \mathcal{N}\left(0, \lambda^2 \left[-t_0 \exp(-\lambda t_0)\right]^2\right)$$

- The asymptotic variance simplifies to $\lambda^2 t_0^2 \exp(-2\lambda t_0)$.

Constructing Confidence Intervals

- We use a plug-in estimator for the standard error. Let $\widehat{\mathsf{SE}}(g(\hat{\theta}))$ be:

$$\widehat{\mathsf{SE}}(g(\hat{\theta})) = \sqrt{\frac{\hat{\sigma}^2 \left[g'(\hat{\theta})\right]^2}{n}}$$

- An approximate $(1-\alpha)100\%$ confidence interval for $g(\theta)$ is:

$$g(\hat{\theta}) \pm z_{\alpha/2} \cdot \widehat{\mathsf{SE}}(g(\hat{\theta}))$$

- Returning to our survival example, $\hat{S}(t_0) = \exp(-\hat{\lambda} t_0)$.
- The estimated standard error is:

$$\widehat{\mathsf{SE}}(\hat{S}(t_0)) = \frac{\hat{\lambda} t_0 \exp(-\hat{\lambda} t_0)}{\sqrt{n}}$$

- A $(1-\alpha)100\%$ asymptotic CI for $S(t_0)$ is:

$$\exp(-\hat{\lambda} t_0) \pm z_{\alpha/2} \left[\frac{\hat{\lambda} t_0 \exp(-\hat{\lambda} t_0)}{\sqrt{n}}\right]$$

Restricting Survival CIs to $[0, 1]$

- The standard linear CI for $\hat{S}(t_0)$ can yield bounds outside $[0, 1]$.
- **Solution:** Apply the Delta Method to a transformation that maps $(0, 1)$ to $(-\infty, \infty)$, such as the complementary log-log transformation: $g(S) = \ln(-\ln(S))$.
- The derivative is $g'(S) = \frac{1}{S\ln(S)}$. By the Delta Method:

$$\widehat{\mathsf{Var}}(\ln(-\ln(\hat{S}))) \approx \left[\frac{1}{\hat{S}\ln(\hat{S})}\right]^2 \widehat{\mathsf{Var}}(\hat{S})$$

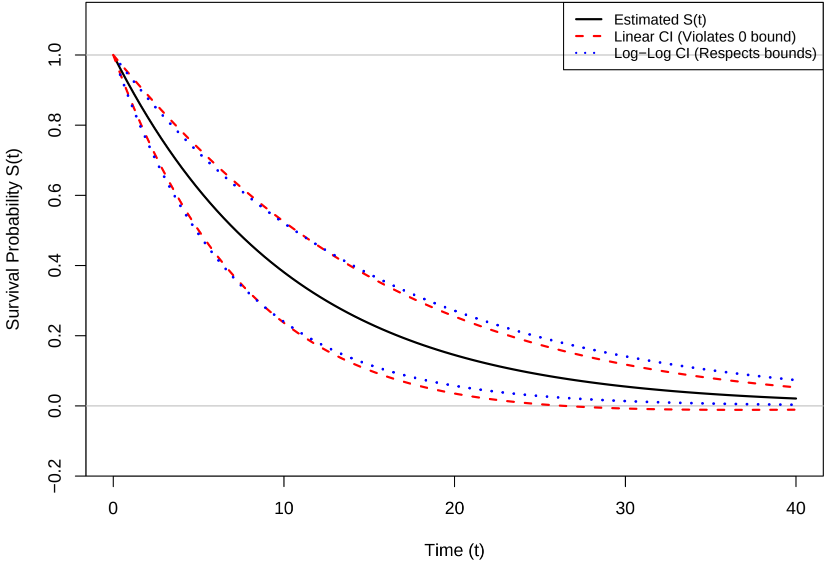
- Calculate the CI bounds for the transformed parameter:

$$[L_W, U_W] = \ln(-\ln(\hat{S})) \pm z_{\alpha/2} \cdot \widehat{\mathsf{SE}}(\ln(-\ln(\hat{S})))$$

- Transform back to get the $[0, 1]$-restricted CI for $S(t_0)$:

$$[\exp(-\exp(U_W)), \exp(-\exp(L_W))]$$

Simulation: Linear vs. Log-Log Bounds



The Second-Order Delta Method

- **Question:** What happens if $g'(\theta) = 0$? The first-order approximation collapses to 0.
- If $g'(\theta) = 0$ but $g''(\theta) \neq 0$, we extend the Taylor series to the second order:

$$g(X_n) \approx g(\theta) + g'(\theta)(X_n - \theta) + \frac{1}{2}g''(\theta)(X_n - \theta)^2$$

- Since $g'(\theta) = 0$, we multiply both sides by n (instead of $\sqrt{n}$):

$$n(g(X_n) - g(\theta)) \approx \frac{1}{2}g''(\theta)\left[\sqrt{n}(X_n - \theta)\right]^2$$

- Because $\sqrt{n}(X_n - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$, its square converges to $\sigma^2\chi^2_1$.
- **Result:** $n(g(X_n) - g(\theta)) \xrightarrow{d} \frac{1}{2}\sigma^2 g''(\theta)\chi^2_1$.

The Multivariate Delta Method

Theorem

Let $\mathbf{X}_n$ be a sequence of random vectors in $\mathbb{R}^k$ such that:

$$\sqrt{n}(\mathbf{X}_n - \theta) \xrightarrow{d} \mathcal{N}_k(\mathbf{0}, \Sigma)$$

Let $g : \mathbb{R}^k \to \mathbb{R}^m$ be a function whose Jacobian matrix $\nabla g(\theta)$ (of size $m \times k$) is evaluated at θ. Then:

$$\sqrt{n}(g(\mathbf{X}_n) - g(\theta)) \xrightarrow{d} \mathcal{N}_m\left(\mathbf{0}, \nabla g(\theta)\Sigma\nabla g(\theta)^T\right)$$

Note: The matrix $\nabla g(\theta)\Sigma\nabla g(\theta)^T$ is the asymptotic covariance matrix of the transformed vector.

Section 2

Bootstrap

What is the Bootstrap?

- The **bootstrap** is a widely applicable and extremely powerful statistical tool.
- It is primarily used to quantify the uncertainty associated with a given estimator.
- **Common use cases:**
  - Estimating the standard errors of an estimator.
  - Constructing confidence intervals of a parameter.
  - Evaluating the performance of a complex machine learning model.
- **Core Idea:** To simulate drawing new samples from a population by resampling using original data.

A Motivating Example: Asset Allocation

- Suppose we wish to invest a fixed sum of money in two financial assets that yield returns X and Y.
- We will invest a fraction α in X and $(1-\alpha)$ in Y.
- **Goal:** Choose α to minimize the total risk, or variance, of our investment.
- The variance of the return is:

$$\text{Var}(\alpha X + (1-\alpha)Y) = \alpha^2\sigma_X^2 + (1-\alpha)^2\sigma_Y^2 + 2\alpha(1-\alpha)\sigma_{XY}$$

- The value that minimizes this risk is:

$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}$$

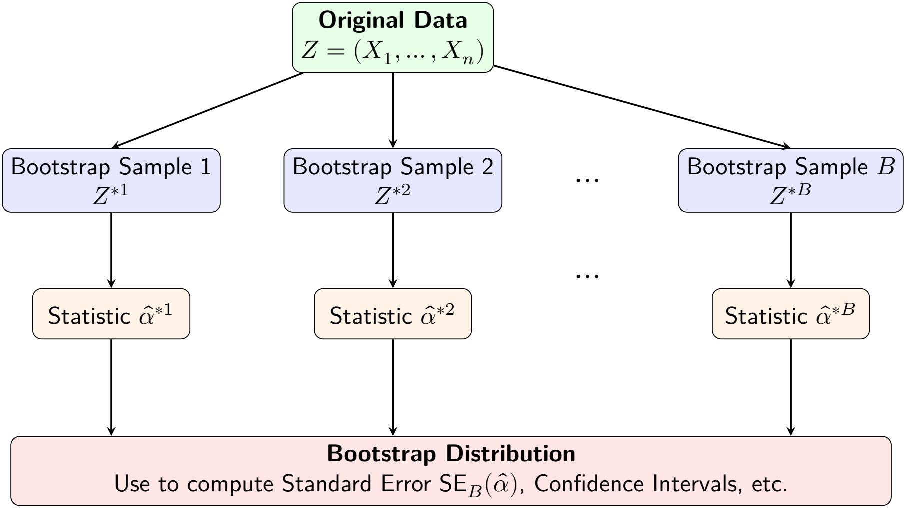
A Motivating Example: Asset Allocation

- σ_X^2, σ_Y^2, and σ_{XY} are generally **unknown**.
- However, we can compute estimates $(\hat{\sigma}_X^2, \hat{\sigma}_Y^2, \hat{\sigma}_{XY})$.
- We then estimate the optimal allocation:

$$\hat{\alpha} = \frac{\hat{\sigma}_Y^2 - \hat{\sigma}_{XY}}{\hat{\sigma}_X^2 + \hat{\sigma}_Y^2 - 2\hat{\sigma}_{XY}}$$

- **Question:** How accurate is $\hat{\alpha}$ as an estimate of α?
- The bootstrap provides a simple way to answer.

The Bootstrap Procedure



Two Flavors of the Bootstrap

1. **Non-Parametric Bootstrap**
   - Makes **no assumptions** about the underlying population distribution.
   - Resamples directly from the observed data (the empirical distribution).
   - **Use when:** The true distribution is unknown, complex, or you cannot confidently commit to a parametric model.
2. **Parametric Bootstrap**
   - Assumes the data comes from a **known parametric family** of distributions (e.g., Normal, Exponential, Poisson).
   - Estimates the parameters from the data, then draws new samples from the fitted theoretical distribution.
   - **Use when:** You have strong theoretical or empirical reasons to assume a specific probability distribution.

Non-Parametric Bootstrap: Example

Goal: Estimate distribution of the sample median.
Observed Data: A sample of $n = 5$: $X = \{3, 8, 2, 10, 5\}$.
Observed Sample Median: 5

The Procedure:

1. **Resample with replacement** directly from X to create B bootstrap datasets of size $n = 5$.
    - Sample $X^{*1} = \{3, 3, 10, 2, 8\} \implies$ Median$^{*1} = 3$
    - Sample $X^{*2} = \{8, 8, 5, 2, 5\} \implies$ Median$^{*2} = 5$
    - Sample $X^{*3} = \{10, 10, 10, 3, 2\} \implies$ Median$^{*3} = 10$
2. **Calculate:** Compute the median for all B (e.g., $B = 1000$) samples. Draw histogram of bootstrap estimates.

Algorithm: The Non-Parametric Bootstrap

Input:

- Observed sample data: $X = (X_1, X_2, \ldots, X_n)$
- Statistic of interest: $\hat{\theta} = T(X)$
- Number of bootstrap replicates: B (typically ≥ 1000)

Procedure:

1. **For** $b = 1$ **to** B **do:**
   - **Step 1 (Resample):** Draw a random sample of size n from X **with replacement** to form the bootstrap sample $X^{*b} = (X_1^{*b}, X_2^{*b}, \ldots, X_n^{*b})$.
   - **Step 2 (Calculate):** Evaluate the statistic on this bootstrap sample:

$$\hat{\theta}^{*b} = T(X^{*b})$$

2. **End For**

Output: $\left\{\hat{\theta}^{*1}, \hat{\theta}^{*2}, \ldots, \hat{\theta}^{*B}\right\}$

Parametric Bootstrap: Example

Goal: Estimate the standard error of the mean time to failure.
Observed Data: A sample of $n = 5$: $x = \{41.1, 14.7, 16.3, 5.6, 22.3\}$.
Probability Model Assumption: The lifetime $X \sim \text{Exp}(\lambda)$.

The Procedure:

1. **Estimate Parameter:** Calculate the maximum likelihood estimate for the rate parameter from the original data, $\hat{\lambda} = 0.05$.
2. **Simulate from Model:** Generate B new datasets of size $n = 5$ directly from $\text{Exp}(\lambda = 0.05)$ using a random number generator.
   - Sample $X^{*1} : \{5.39, 54.23, 16.40, 38.47, 3.89\} \implies \bar{X}^{*1} = 23.67$
   - Sample $X^{*2} : \{20.88, 7.14, 62.54, 62.50, 5.98\} \implies \bar{X}^{*2} = 31.81$
   - Sample $X^{*3} : \{1.40, 45.69, 2.19, 11.02, 41.20\} \implies \bar{X}^{*2} = 20.30$
3. **Calculate:** Compute the mean for all B (e.g., $B = 1000$) samples. Draw histogram of bootstrap estimates.

Algorithm: The Parametric Bootstrap

Input:

- Observed sample data: $X = (X_1, X_2, \dots, X_n)$
- Assumed parametric probability model: F_θ
- Statistic of interest: $\hat{\alpha} = T(X)$
- Number of bootstrap replicates: B (typically ≥ 1000)

Algorithm: The Parametric Bootstrap (Cont.)

Procedure:

1. **Estimate:** Calculate the parameter estimate $\widehat{\theta}$ from the original sample X (e.g., via Maximum Likelihood Estimation).
2. **For** $b = 1$ **to** B **do:**
   - **Step 1 (Simulate):** Generate a random sample of size n directly from the fitted distribution $F_{\hat{\theta}}$ to form the bootstrap sample $X^{*b} = (X_1^{*b}, X_2^{*b}, \dots, X_n^{*b})$.
   - **Step 2 (Calculate):** Evaluate the statistic on this simulated bootstrap sample:
   $$\widehat{\theta}^{*b} = T(X^{*b})$$
3. **End For**

Output: $\left\{\widehat{\theta}^{*1}, \widehat{\theta}^{*2}, \dots, \widehat{\theta}^{*B}\right\}$

Estimating the Standard Error

- Once we have our B bootstrap estimates, we can analyze their distribution.
- Bootstrap estimates : $\hat{\alpha}^{*1}$, $\hat{\alpha}^{*2}$, ... , $\hat{\alpha}^{*B}$.
- The **bootstrap standard error** is simply the sample standard deviation of the bootstrap estimates:

$$\mathsf{SE}_B(\hat{\alpha}) = \sqrt{\frac{1}{B-1}\sum_{i=1}^{B}\left(\hat{\alpha}^{*i} - \frac{1}{B}\sum_{i'=1}^{B}\hat{\alpha}^{*i'}\right)^2}$$

- This serves as an estimate of the standard error of $\hat{\alpha}$ estimated from the original data set.

Bootstrap Confidence Intervals: Overview

- While the Bootstrap Standard Error (SE_B) provides a measure of spread, we often want a formal Confidence Interval (CI) for our parameter θ.
- If the estimator $\hat{\theta}$ is strictly normally distributed, we could simply use the Normal Approximation:

$$\hat{\theta} \pm z_{1-\alpha/2} \cdot \mathsf{SE}_B(\hat{\theta})$$

- However, the primary power of the bootstrap is that we **do not need to rely on the normality assumption**.
- We can compute intervals directly from the empirical bootstrap distribution.

The Percentile Bootstrap CI

- The **Percentile Interval** is the most intuitive approach.
- We simply use the empirical quantiles of our computed bootstrap distribution.
- Let $\widehat{\theta}^*_{(\alpha/2)}$ and $\widehat{\theta}^*_{(1-\alpha/2)}$ be the $\alpha/2$ and $1-\alpha/2$ quantiles of the B bootstrap estimates.
- The $100(1-\alpha)\%$ Percentile Bootstrap CI is:

$$\left[\widehat{\theta}^*_{(\alpha/2)},\ \widehat{\theta}^*_{(1-\alpha/2)}\right]$$

- **Advantage:** Very simple to understand and compute; respects the natural mathematical bounds of the parameter.

The Basic (Pivotal) Bootstrap CI

- The **Basic Interval** relies on approximating the distribution of the pivotal quantity $\delta = \widehat{\theta} - \theta$.
- We approximate the true distribution of R using the bootstrap distribution of $\delta^* = \widehat{\theta}^* - \widehat{\theta}$.
- Substituting the bootstrap quantiles and rearranging to isolate θ yields the Basic CI:

$$\left[2\widehat{\theta} - \widehat{\theta}^*_{(1-\alpha/2)},\ 2\widehat{\theta} - \widehat{\theta}^*_{(\alpha/2)}\right]$$

An empirical bootstrap confidence interval

- The sample data: 30, 37, 36, 43, 42, 43, 43, 46, 41, 42
- Problem: Estimate the mean μ of the underlying distribution and give an 80% bootstrap confidence interval.
- The sample mean is $\bar{x} = 40.3$. We use this as an estimate of the true mean μ of the underlying distribution.
- To make the confidence interval we need to know how much the distribution of $\bar{x}$ varies around μ.
- In fact, we want to know the distribution of $\delta = \bar{x} - \mu$.
- It says that we can approximate it by the distribution of $\delta^* = \bar{x}^* - \bar{x}$ where $\bar{x}^*$ is the mean of an empirical bootstrap sample.

An empirical bootstrap confidence interval (cont.)

Bootstrap Samples (20 iterations):

43	36	46	30	43	43	43	37	42	42	43	37	36	42	43	43	42	43	42	43
43	41	37	37	43	43	46	36	41	43	43	42	41	43	46	36	43	43	43	42
42	43	37	43	46	37	36	41	36	43	41	36	37	30	46	46	42	36	36	43
37	42	43	41	41	42	36	42	42	43	42	43	41	43	36	43	43	41	42	46
42	36	43	43	42	37	42	42	42	46	30	43	36	43	43	42	37	36	42	30
36	36	42	42	36	36	43	41	30	42	37	43	41	41	43	43	42	46	43	37
43	37	41	43	41	42	43	46	46	36	43	42	43	30	41	46	43	46	30	43
41	42	30	42	37	43	43	42	43	43	46	43	30	42	30	42	30	43	43	42
46	42	42	43	41	42	30	37	30	42	43	42	43	37	37	37	42	43	43	46
42	43	43	41	42	36	43	30	37	43	42	43	41	36	37	41	43	42	43	43

An empirical bootstrap confidence interval (cont.)

- Next we compute $\delta^* = \bar{x}^* - \bar{x}$ for each bootstrap sample and sort them smallest to biggest:

-1.6	-1.4	-1.4	-0.9	-0.5	-0.2	-0.1	0.1	0.2	0.2
0.4	0.4	0.7	0.9	1.1	1.2	1.2	1.6	1.6	2.0

- 90th percentile -> 18th element -> 1.6
- 10th percentile -> 2nd element -> -1.4
- Therefore our bootstrap 80% confidence interval for μ is

$$[\bar{x} - \delta^*_{0.1}, \bar{x} - \delta^*_{0.9}] = [40.3 - 1.6, 40.3 + 1.4] = [38.7, 41.7]$$

- In this example we only generated 20 bootstrap samples so they would fit on the page.

References

- **James, G., Witten, D., Hastie, T., & Tibshirani, R.** (2013). *An Introduction to Statistical Learning with Applications in R*. Springer.
- **StatMethods.** *Bootstrapping: Quick-R*. statmethods.net/advstats/bootstrapping.html